Ram Calculus Certificates for Thermal FK–Ising Interfaces and Massive SLE

Boundary-Correct Reformulation, Residual Bounds, and the Moving-Domain Boundary

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Abstract

We examine the complete pinned source of *Thermal FK–Ising interfaces and massive SLE*, with particular attention to its nonlinear scalar equation $\Delta h = -4m|\nabla h|$, discontinuous prime-end boundary data, Green potential, and stochastic driving coefficient. Actual passages and proofs from the Ram PDE-first, Urysohn, stability, and compatible-panel reports are matched to specific proof obligations. We rederive the boundary-aware Green representation and prove a bounded-residual certificate on rough simply connected domains without a small-mass condition. On fixed Lipschitz domains with a valid trace lift we prove contraction, nonlinear flux, mass-perturbation, and regularization estimates. We also give a conditional smooth-domain comparison and an explicit compact-drift certificate with an indispensable tail budget. The gradient norm is not treated as a smooth nonlinearity at zero, and a discontinuous bank value is not silently lifted into H^1. The source's curve compactness, moving-mark energy, simultaneous bracket calculus, and remote-domain shielding remain separate arguments; none is certified by the finite diagnostics here. Contribution assessment: C2/5, explicit restricted residual and perturbation certificates are obtained, but no new interface limit, source-wide proof certification, or global improvement is established.

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1 Assignment, provenance, and claim discipline

The assigned manuscript is M419, family 223, catalogue applicability **L4**. Its public source is OpenAI's October 5, 2026 manuscript [1], at commit

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

The existing local upstream audit of October 8 reports that the selected 100 sources, hence M419, are unchanged at

`fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`.

That audit is used as provenance, not repeated or interpreted as a proof check. The directory is

`Thermal-FK-Ising-interfaces-and-massive-SLE-October-5-2026/build`.

Every line of `main.tex`, `references.tex`, and all seven section files was read. The counts are 69, 65, 112 (introduction), 266 (model), 727 (geometry), 594 (energy), 992 (calculus), 222 (uniqueness), and 896 (localization), respectively: 3,943 source lines in all. This includes the proofs, remarks, two diagram definitions, and bibliography, not only statements or a screening map. Counts and hashes are recorded separately.

The review is a *bounded mathematical audit*: it follows the proof dependencies, checks central normalizations and deductions, and identifies the precise assumptions of the extensions proved below. It is not formal verification, a replay of lattice probability estimates, or independent certification of every external theorem. No definite contradiction in the assigned source is established here. In particular, the boundary-lift and derivative obstacles below obstruct an indiscriminate Ram transfer, not the source's actual bounded-solution argument.

Three evidence categories will be kept distinct:

1. **Source assertions and rederivations**: inherited mechanisms, attributed by file and label, with dependencies retained.
2. **Proved restricted certificates**: full proofs in this report, whose settings may be narrower than the source's main theorem.
3. **Unverified inputs or future work**: especially tail enclosures, continuum inverse norms, moving-domain control, and stochastic law transfer.

The private corpus is confidential local evidence. It was not uploaded, and no private text or private path was used in an Internet query.

2 The exact source problem

2.1 Thermal weights and what the main theorem asserts

The source takes a bounded simply connected physical domain Ω, an interior normalization point, distinct boundary prime ends a, b, and a nonzero real mass. Rhombic isoradial approximants have half-angles in $[\eta, \pi/2 - \eta]$ for a fixed $\eta > 0$. The nome satisfies $r_\delta/\delta \to |m|/2$. For positive mass the abstract edge angle $\widehat{\theta}$ is specified by

$$\int_0^{\widehat{\theta}} (1 - k_\delta^2 \sin^2 s)^{-1/2} \, \mathrm{d}s = \frac{2K(k_\delta)}{\pi}\theta, \qquad r_\delta = e^{-\pi K(\sqrt{1-k_\delta^2})/K(k_\delta)}.$$

Negative mass uses the complementary angle on the dual lattice. The loop weight is $(\sqrt{2})^{\ell(E)}$, with edge factors $\sin(\widehat{\theta}_e/2)$ for primal-open edges and $\sin(\pi/4 - \widehat{\theta}_e/2)$ for their complements. The FK cluster weight is two, not the nome.

The source's `thm:main` asserts full-sequence convergence for marked Carathéodory approximants, even with diverging diameters. Curves are transferred by normalized conformal maps to $\overline{\mathbb{D}}$ and compared uniformly modulo increasing reparametrization. This does not assert a continuous Euclidean boundary parametrization of an arbitrary rough physical domain. Negative mass is treated by duality and reversal, not by merely substituting a negative coefficient into a positive-mass stochastic estimate.

2.2 Scalar equation and exact conformal factors

With $D_z = 2i\partial_z$, the source uses

$$D_z h = f^2, \qquad \bar{\partial} f = -im\bar{f}, \qquad |\nabla h| = |f|^2, \qquad \Delta h = -4m|\nabla h|. \tag{2.1}$$

Indeed $D_z h = h_y + ih_x$ and $\bar{\partial} D_z h = (i/2)\Delta h$, whereas $\bar{\partial}(f^2) = -2im|f|^2$. These identities verify the sign and factor four. The solution class in `model.tex`, `mod:bvp`, is bounded, classical in the interior, with prime-end limits one and zero on the two open banks. The values at the marked ends

are not prescribed. The source's existence statement imports the two-point observable; bounded scalar uniqueness is proved separately.

For a continuous Loewner driver W, set

$$\begin{aligned} \partial_t g_t(z) &= \frac{2}{g_t(z) - W_t}, \quad Z_t = g_t \circ \phi - W_t, \quad \psi_t = Z_t^{-1}, \\ \mu_t(v) &= m|\psi_t'(v)|, \quad P(v) = \Im(-1/v) = \frac{\Im v}{|v|^2}. \end{aligned} \tag{2.2}$$

If $H_t = h_t \circ \psi_t$, the conformal chain rule gives

$$\Delta_v H_t = -4\mu_t |\nabla_v H_t|, \qquad |\nabla_v H_t| = |\psi_t'|\,|\nabla h_t| \circ \psi_t. \tag{2.3}$$

The mass in the chart is variable, not the physical constant m. The source's driving coefficient and SDE are

$$C_t = \frac{4}{\pi} \int_{\mathbb{H}} P\mu_t |\nabla H_t| \,\mathrm{d}A, \qquad \mathrm{d}W_t = \sqrt{16/3}\,\mathrm{d}B_t + \frac{2\pi}{3} C_t \,\mathrm{d}t. \tag{2.4}$$

Its law definition requires the actual integral to be finite almost everywhere, $\int_0^T C_t^2 \,\mathrm{d}t < \infty$ for finite T, and the full capacity-parametrized trace. A dummy value for an undefined integral does not remove those requirements. Scaling the initial chart by $\lambda > 0$ sends t, W, C to $\lambda^2 t, \lambda W, C/\lambda$, which checks both the SDE and its energy convention.

3 Source-wide dependency audit

The following ledger covers every substantive section. "Read" means the displayed source proofs were inspected in full, not that their external inputs have all been independently proved.

3.1 Model and geometry

`mod:bvp-unique` subtracts two solutions and uses

$$\beta = 4m \frac{\nabla h_1 + \nabla h_2}{|\nabla h_1| + |\nabla h_2|}, \qquad |\beta| \le 4|m|, \tag{3.1}$$

with zero at a simultaneous zero of the gradients. The difference solves $\Delta q + \beta \cdot \nabla q = 0$. A bounded-drift killed diffusion and nonatomic prime-end harmonic measure remove the two exceptional ends. No H^1 lift or derivative of the norm is used. Domain continuity is a diagonal consequence of the observable theorem on uniformly physically bounded domains; it is not a quantitative Lipschitz estimate in a slit.

In `geometry.tex`, `prop:inputs` imports the observable and crossing results. `geo:domain-markov` retains incident sectors at pinches and proves uniform positivity of genuine local decisions. `geo:arm` compares the union of one-color-bank regions at once; artificial boundary identifications are not counted as open arm edges. `geo:g2` distinguishes forced from unforced components and deals with the submesh drawing convention. These are essential probabilistic and topological obligations, not consequences of a PDE residual.

`geo:crosscut` uses total image length and length–area on an annulus to obtain a logarithmic modulus for connected interior sets; endpoints must retain their incident prime-end occurrences. `geo:compactness` then invokes the conditional crossing-to-Loewner framework. `geo:active` identifies both the active kernel and the moving prime end by a short actual continuation chain. Kernel convergence alone is not substituted for marked convergence.

`geo:restart` transfers conditional estimates with protected coordinates and relaxed thresholds. `geo:martingale` lifts squares to signed martingales: both genuine next decisions have probability at least p_0, preventing a macroscopic sign flip away from zeros; near zeros the amplitude is small. Its martingale assertion is only before swallowing. `geo:filtration` correctly distinguishes the natural driver filtration used for crossing estimates from the joint field filtration used later. These limiting and stopping arguments were read; no independent discrete simulation or proof of their probability estimates is claimed.

3.2 Moving-mark energy: a rederivation of the analytic assembly

For a live physical point w, write $Z_t = r_t e^{i\theta_t}$, $Y_t = \Im Z_t$, $X_t = Y_t/r_t$, and $s = \int_0^t r_u^{-2}\,\mathrm{d}u$. The imaginary Loewner equation, which does not differentiate W, gives

$$Y_s = Y_0 e^{-2s}, \qquad \mathrm{d}\log \mathrm{crad}_{O_{t(s)}}(w) = -4X_s^2\,\mathrm{d}s, \qquad \mathrm{d}t = r_s^2\,\mathrm{d}s. \tag{3.2}$$

The source kills X at its clock lifetime, not the conformal radius before swallowing. `eng:restarts` supplies boundary-visit and bulk conformal-radius tails. `eng:reflection` reflects across an avoided real interval and integrates the fixed-sign identity

$$\partial_t(l_t''/l_t')(x) = 4l_t'(x)/(l_t(x) - W_t)^3.$$

This controls both the supremum and occupation of X_s/X_0 on avoidance. `eng:controls` joins that estimate to the crossing tails; `eng:decay` obtains $\mathbb{E}X_s^p \le KX_0^p e^{-bs}$ for a small p, not an unjustified contraction of every moment.

The key bookkeeping in `eng:weighted` retains both ratios $r_s/\sqrt{t(s)}$ and its inverse. For the running-minimum bins $2^{-k-1}X_0 < \min_{v\le s} X_v \le 2^{-k}X_0$, it interpolates $KX_0^{2p}2^{-2pk}$ with $KX_0^p e^{-bs}$, obtaining $KX_0^{3p/2}2^{-pk}e^{-bs/2}$. Thus $0 < \epsilon < \min(p, b/4, 2)$ gives

$$\mathbb{E}\int X_s^{2p}\left[(\sqrt{t(s)}/r_s)^\epsilon + (r_s/\sqrt{t(s)})^\epsilon\right]\,\mathrm{d}s \le K.$$

Finally, on shells $|v| \asymp 2^j\sqrt{t}$, Cauchy–Schwarz has angular factor $\int_0^\pi \sin^{-2p}\theta\,\mathrm{d}\theta < \infty$. Summing with $2^{-\epsilon|j|}$ and changing area by $\mu_t^2\,\mathrm{d}A(v) = m^2\,\mathrm{d}A(w)$ yields

$$J_t = \int_{\mathbb{H}} \frac{\mu_t(v)}{|v|^2}\,\mathrm{d}A(v), \qquad \mathbb{E}\int_0^\infty J_t^2\,\mathrm{d}t \le Km^2|\Omega|. \tag{3.3}$$

This is the assembly of `eng:energy`, conditional on the source's geometric lemmas. The killed integrands and Tonelli, rather than a simultaneous exceptional set for all w, justify the area integration. None of our fixed-domain estimates supplies these restarted tails.

3.3 Green calculus and stochastic identification

`cal:potential` proves the representation rederived in Section 5. `cal:stochastic-potential` inserts a chart height cutoff $\chi(Y_t/\lambda)$ before swallowing. Its loss is bounded by $K\int_{\Im v\le 2\lambda}\mu_t|v|^{-2}\,\mathrm{d}A\,\mathrm{d}t$; the expectation vanishes using (3.3). A zero extension of a martingale through swallowing is not presumed. Weighted covariance Cauchy–Schwarz controls spatial stochastic integration.

`cal:recovery` recovers the continuous driver locally from the difference of harmonic measures at two protected points. The derivative in the driver is $(P_t(z) - P_t(z'))/\pi$; rational discs, finitely many pairs, and nonaccumulating chart changes give global stopping localizers. This avoids assuming semimartingality at the beginning. `cal:covariance` builds a simultaneous clock dominating

capacity, driver variation, driver bracket, and spatial bracket measures. Spatial elliptic estimates make evaluation bounded on the stopped massive-holomorphic Hilbert spaces. Separate pointwise null sets would not suffice for later boundary-approaching balls.

Here is the algebraic core of `cal:driver`, with the analytic requirements visible. At a time in that common full set, let $\tau = \mathrm{d}t/\,\mathrm{d}L$, $\nu = \mathrm{d}V/\,\mathrm{d}L$, U be the driver's covariance vector, and $A(z)$ the fermion's covariance vector in the current chart. The source establishes

$$F^2 = (\pi z)^{-1} + 4\int DG\,\mu|F|^2, \tag{3.4}$$

$$2FA = U/(\pi z^2) + 8\int DG\,\mu\Re(\bar{F}A), \tag{3.5}$$

$$A\cdot A = (|U|^2 - 4\tau)/(\pi z^3) + \nu/(\pi z^2) - 2C\tau/z^2 + 4\int DG\,\mu|A|^2. \tag{3.6}$$

The dot product in the last identity is bilinear after complexification. In `cal:tip-expansions`, $\mu \in L^2$ and the finite weighted bracket measure yield $F^2 = (\pi z)^{-1} + o(s^{-1})$ uniformly on $B(is, s/4)$, mixed remainder $o(s^{-3/2})$ in normalized L^2, and bracket-potential remainder $o(s^{-2})$ in normalized L^1. Squaring (3.5), dividing by $4F^2$, and comparing cubic terms with (3.6) gives

$$|U|^2/4 = |U|^2 - 4\tau, \qquad |U|^2 = (16/3)\tau.$$

When $\tau = 0$, the mixed square is $o(s^{-2})$, so $\nu = 0$ without requiring $J < \infty$ on a capacity-singular set. When $\tau > 0$, the energy estimate permits the refined expansion $F^2 = (\pi z)^{-1} + C + o(1)$. The quadratic comparison is

$$\nu/\pi - 2C\tau = -|U|^2 C/4, \qquad \nu = (2\pi/3)C\tau.$$

Testing after $z = s\zeta$ makes these averaged coefficient comparisons legitimate. This identifies (2.4) only after all the preceding spatial, stochastic, and null-set controls. Ram coordinate algebra does not replace them.

3.4 Uniqueness and the unbounded-approximant reduction

In `uni:measurable`, compact exhaustion makes C a progressive, nonanticipating functional of the stopped driver. `uni:weak` proves weak uniqueness in the finite-energy class, not pathwise uniqueness. For drift b, capacity horizon T, and $\sigma_n = \inf\{t : \int_0^t b^2 \geq n\} \wedge T$, a candidate solution attains its energy level continuously. Stopped Girsanov removes its drift. The resulting full Wiener path agrees up to σ_n, so nonanticipation identifies the same stopping rule under the comparison law. The inverse density is consequently a fixed Wiener functional. The stopped laws agree and $\sigma_n = T$ eventually under each candidate. This argument avoids assuming continuous energy on every arbitrary Wiener path. `uni:bounded` combines it with curve compactness and driver identification, retaining the full trace requirement.

All of `localization.tex` was read. `loc:remote` uses restricted length–area inside a fixed physical ball, not a bound on approximant area. `loc:obstructions` controls maps after small attached changes. `loc:gates` organizes mandatory visible gates in a tree; `loc:heads` protects unqueried endpoint access. `loc:shield` explores opposite-color clusters from exterior seeds, leaving a monochromatic frontier and an ordinary conditional Dobrushin law on success. Success is determined by the complete revelation record, not an extra conditioning on unqueried edges.

`loc:failure` bounds failure by finite-head terms and products $\prod_j K(2^{-j}(j+2)^2)^\alpha$. Their logarithm is $-(\alpha \log 2)k^2/2 + O(k\log(k+2))$, which dominates both gate counts and the δ^{-2}

final-cell factor. `loc:reduction` uses staged subsequences and uniform successful-output control; `loc:universality` integrates over successful records and invokes a subsequence criterion. The final duality paragraph exchanges primal/dual labels and reverses orientation. These graph, topology, and measure arguments are retained as source claims; the present certificates neither repair a demonstrated gap there nor independently certify them.

4 Actual Ram passages and the transfer boundary

All three local maps `corpus-core-map.txt`, `corpus-foundations-map.txt`, and `corpus-applications-map.txt` were read as routing aids. The following original passages, including their proofs, were then read directly. Their source numbering belongs to the named series; bare "TR8" from another series is not interchangeable.

Private source	Actual locator and mechanism	Use and retained restriction
Astra TR8 [8]	`sec:elliptic`, `thm:coercive`, `eq:operator-domain`, `thm:green-equivalence`; lines 53–395 read.	Lax–Milgram inverse $V' \to V$, lift correction, actual Dirichlet Green operator. Requires admissible trace and source space.
Astra TR8	`sec:semilinear`, `eq:newton-full`; compatible panels and `eq:weak-residual`, lines 492–590 read.	Derivative-dependent nonlinearity must map the chosen spaces. Finite residual rows are not a continuum dual-norm upper bound.
Astra TR6 [9]	`eq:localidentity`, full Urysohn variation `eq:du`, `eq:ddu`, `eq:newtonfull`; lines 269–427 read.	Exact displacement localization and full state derivative. Derivatives inside the state require a stronger domain than C.
Astra TR6	`eq:shapevolume`, `eq:shapeboundary`, ball-local solvability; lines 565–695. Polynomial/moment discussion, lines 872–947.	Smooth diffeomorphic domain transport only; nonlinear composition need not preserve polynomial degree.
Astra TR1 [10]	`stability.tex`, full file: `thm:pert`, `thm:ball`, `eq:radii`, `prop:transfer`.	Physical scaling, residual balls, full Banach isomorphism, and explicit error budgets. No rectangular left-inverse root inference.
Foundation revised old-TR250 [11]	Lines 155–178, 455–581: `eq:moments`, polynomial construction, `thm:hypercircle`, `eq:source-est`.	Unisolvent finite moments, conforming reconstruction and equilibrated $H(\mathrm{div})$ flux. Prager–Synge is explicitly classical.

No claim is made to have read all four private reports or the entire corpus. TR1's selected file is complete; the other scopes are exactly the stated ranges (with the recorded short overlaps). The proofs above use established contraction, elliptic, and flux arguments. Their Ram role is a domain-correct representation and a transparent certificate, not priority for those arguments.

4.1 Why the bank jump is not an H^1 lifting datum

Proposition 4.1. *Even on a smooth planar domain, a boundary function with a nonzero jump between adjacent open arcs need not be in $H^{1/2}(\partial D)$ and cannot then be the trace of an $H^1(D)$ function.*

Proof. Use a smooth boundary coordinate s across a marked point, with values zero for $s > 0$ and one for $s < 0$. The local fractional seminorm contains, up to fixed positive coordinate factors,

$$\int_0^a \int_0^a \frac{1}{(s+t)^2}\,\mathrm{d}s\,\mathrm{d}t = \infty.$$

Indeed the inner integral is $1/t - 1/(a+t)$ and its integral at zero diverges. The trace theorem on a Lipschitz domain rules out an H^1 lift. Equivalently, on a half-annulus the harmonic jump profile $h = \arg z/\pi$ has $|\nabla h| = 1/(\pi r)$ and energy $\pi^{-1}\log(R/\rho)$, diverging as $\rho \downarrow 0$. □

This does not prove that every difference of two singular solutions has infinite energy. A singularity-subtracted difference may have finite energy, but that is an additional theorem, not a free license to apply TR8 to the original h. Smoothing the bank transition or cutting out the marks changes the boundary problem and creates a boundary-error budget. On rough prime-end boundaries a classical trace theorem is even less automatic.

4.2 The other obstruction: differentiating the gradient norm

The pointwise map $p \mapsto |p|$ is 1-Lipschitz but has no linear derivative at $p = 0$. More concretely, at a constant field, $|\nabla(c + tw)| = |t|\,|\nabla w|$, whose two-sided quotient is not a linear function of w. Thus an unrestricted Fréchet-Newton theorem on that neighborhood is false. Nonvanishing gradients on a controlled C^1 neighborhood or a positive regularization parameter permits differentiation; H^1 control alone does not guarantee either.

5 Green reformulation with the boundary retained

For $m > 0$, a source solution h and its harmonic bank function $u(z) = \arg Z(z)/\pi$ satisfy

$$h(z) = u(z) + 4m \int_O G_O(z,w)|\nabla h(w)|\,\mathrm{d}A(w), \quad G_{\mathbb{H}}(z,v) = \frac{1}{2\pi}\log\left|\frac{z-\bar{v}}{z-v}\right|. \tag{5.1}$$

Here G_O is for $-\Delta$, not the free-space fundamental solution. To rederive (5.1) without a global Sobolev lift, exhaust O by images of concentric discs in an interior Riemann chart. Green's formula for $h - u$ on each exhaustion domain gives its harmonic boundary extension plus the nonnegative potential. The boundary extension tends to zero by bounded convergence in disc Poisson measure: the two marks have measure zero and all other prime-end limits vanish. Increasing Green kernels and monotone convergence give (5.1). As $0 \le h, u \le 1$, its potential obeys

$$0 \le \int_O G_O(z,w)|\nabla h(w)|\,\mathrm{d}A(w) \le (4m)^{-1}. \tag{5.2}$$

This is the source's `cal:potential` mechanism; it is a different route from TR8's H^1 lifting theorem.

Set $K(u,a) = G_O(u+a,u)$ and $q(w) = |\nabla h(w)|$. The precise Ram-localized identity from TR8 and TR6 is

$$h(x) - u(x) = 4m \int_{x-O} K(x-a,a)q(x-a)\,\mathrm{d}A(a). \tag{5.3}$$

The displacement set is $x - O$ and the absolute Jacobian is one. It retains the diagonal singularity, the full support, and the boundary condition. It is not a terminating local differential inverse.

Proposition 5.1 (A legitimate moment replacement)**.** *Fix an observation x and a measurable source panel $E \subset O$. Suppose q has a C^{N+1} extension to all segments from a chosen center c to E, with multilinear derivative norm at most M. Then*

$$\int_E G_O(x,w)q(w)\,\mathrm{d}A(w) = \sum_{|\alpha|\le N} \frac{\partial^\alpha q(c)}{\alpha!} \int_E G_O(x,w)(w-c)^\alpha\,\mathrm{d}A(w) + R_N(x), \tag{5.4}$$

$$|R_N(x)| \le \frac{M}{(N+1)!} \int_E |G_O(x,w)|\,|w-c|^{N+1}\,\mathrm{d}A(w). \tag{5.5}$$

The formula is exact if q is in the declared polynomial class and the moments are exact. Any omitted source set contributes its own integral.

Proof. Taylor's formula along $c + t(w-c)$ gives the multi-index polynomial and remainder bounded by $M|w-c|^{N+1}/(N+1)!$. Multiply by the Green kernel and integrate the absolute remainder. The finite sum commutes with integration. A logarithmic diagonal singularity is locally integrable against the bounded polynomial factors, so it does not invalidate the formula when the displayed absolute moment is finite. □

Polynomial h does not imply polynomial $|\nabla h|$. At a gradient zero the assumed regularity of q may fail, so even an exact Taylor jet of h does not justify (5.4). Singularity-adapted integration, verified quadrature, or weak residuals are alternatives.

For C_t, ordinary Green moments are not enough: differentiating at a boundary mark replaces the logarithmic kernel by a Poisson-type weight. The source obtains its constant term only after proving $J_t < \infty$. That weighted condition cannot be read off from a small interior potential error in (5.2).

6 An all-mass bounded-residual certificate

The source's comparison proof has a useful quantitative consequence which does not require a global H^1 lift or differentiability of the norm. It is a classical bounded-drift barrier argument, specialized here to the actual nonlinear equation.

Theorem 6.1 (Rough-domain sup-norm certificate)**.** *Let $O \subset \mathbb{C}$ be bounded and simply connected, contained after a rigid motion in the strip $|x_1| < R$. Let $m \in \mathbb{R}$. Suppose h, v are bounded real $C^2(O)$ functions, $\Delta h + 4m|\nabla h| = 0$, and*

$$r_v = \Delta v + 4m|\nabla v| \in L^\infty(O).$$

Assume $h - v$ has prime-end limits of absolute value at most d off a finite set of specified prime ends. For any $k > 4|m|$, define

$$B_{R,m,k} = \frac{\cosh(kR) - 1}{k(k - 4|m|)}. \tag{6.1}$$

Then

$$\|h - v\|_{L^\infty(O)} \le d + B_{R,m,k}\,\|r_v\|_\infty\,. \tag{6.2}$$

This is an error estimate relative to the stipulated solution h, not an existence theorem for an arbitrary approximate field.

Proof. Put $e = h - v$ and use (3.1) with $h_1 = h$, $h_2 = v$. Then $\mathcal{L}e = -r_v$, where $\mathcal{L} = \Delta + \beta \cdot \nabla$ and $|\beta| \le b = 4|m|$. Extend the drift boundedly to the plane. Bounded-drift Girsanov on each finite time interval constructs a weak diffusion $\mathrm{d}X = \sqrt{2}\,\mathrm{d}B + \beta(X)\,\mathrm{d}t$ and makes its path law equivalent to Brownian law there. Let τ be exit from O.

The smooth nonnegative strip barrier

$$w(x) = \frac{\cosh(kR) - \cosh(kx_1)}{k(k-b)}$$

satisfies $0 \le w \le B_{R,m,k}$ on O and

$$-\mathcal{L}w = \frac{k\cosh(kx_1) + \beta_1 \sinh(kx_1)}{k-b} \ge \frac{k\cosh(kx_1) - b|\sinh(kx_1)|}{k-b} \ge 1.$$

Apply Itô's formula up to a compact exhaustion exit and a finite horizon. Nonnegativity of w gives the expected stopped lifetime at most $w(x)$. Monotone exhaustion and then increasing horizons yield $\mathbb{E}_x\tau \le w(x) \le B_{R,m,k}$, and $\tau < \infty$ almost surely.

Brownian exit has a prime-end limit and its harmonic measure is nonatomic. Finite-horizon equivalence, followed by the countable union over horizons containing τ, transfers these facts to X. In particular the finite exceptional set is not hit as an exit prime end. Itô's formula for e on compact exhaustions gives

$$e(x) = \mathbb{E}_x e(X_\tau) + \mathbb{E}_x \int_0^\tau r_v(X_s)\,\mathrm{d}s,$$

where the first term denotes the limiting prime-end value. Boundedness of e, the assumed boundary limits, and the integrable lifetime justify both limits by dominated convergence. Its absolute value is at most $d + \|r_v\|_\infty \mathbb{E}_x\tau$, proving the result. □

Corollary 6.2 (Conditional regularization budget)**.** *Under the boundary and boundedness hypotheses of Theorem 6.1, if v solves $\Delta v = -4m\sqrt{|\nabla v|^2 + \varepsilon^2}$ with $\varepsilon > 0$, then*

$$\|h - v\|_\infty \le d + 4|m|B_{R,m,k}\varepsilon.$$

Proof. For $a \ge 0$, $0 \le \sqrt{a^2+\varepsilon^2} - a \le \varepsilon$, so the unregularized residual of v is at most $4|m|\varepsilon$. Apply the theorem. □

No rough-domain regularized solution is constructed by this corollary. Its global residual requirement can be demanding near marks, and the strip constant can grow rapidly with $|m|R$. A local protected subdomain permits a smaller constant but needs a certified boundary discrepancy there. A small sup-norm error is not a gradient or drift-error bound.

7 Energy and flux certificates with valid lifts

Here the setting changes deliberately. Let $D \subset \mathbb{R}^2$ be bounded, connected, and Lipschitz. All functions are real. Set $V = H_0^1(D)$ with $\|w\|_V = \|\nabla w\|_2$, and fix a known Poincaré bound

$$\|w\|_2 \le C_P \|\nabla w\|_2 \quad (w \in V). \tag{7.1}$$

Let $b \in H^{1/2}(\partial D)$ and $\ell \in H^1(D)$ be an actual lift. Let $L : V \to V'$ be the Riesz operator $\langle Lw, \phi\rangle = \int_D \nabla w \cdot \nabla\phi$ and $G = L^{-1}$. TR8's coercive proof gives $\|G\|_{V'\to V} = 1$ and $\|Gf\|_V \le C_P \|f\|_2$ for $f \in L^2$. The harmonic lift is $H = \ell - G(L\ell)$, where $L\ell$ is defined weakly against V.

Theorem 7.1 (Fixed-domain contraction and residual)**.** *If $q = 4|m|C_P < 1$, there is exactly one $u \in \ell + V$ satisfying*

$$\int_D \nabla u \cdot \nabla \phi = 4m \int_D |\nabla u| \phi \quad (\phi \in V). \tag{7.2}$$

For any $v \in \ell + V$ define its full weak residual

$$\langle \mathcal{R}_m(v), \phi \rangle = \int_D \nabla v \cdot \nabla \phi - 4m \int_D |\nabla v| \phi. \tag{7.3}$$

Then

$$\|\nabla(u - v)\|_2 \le \frac{\|\mathcal{R}_m(v)\|_{V'}}{1 - q}, \qquad \|\nabla u\|_2 \le \frac{\|\nabla H\|_2}{\sqrt{1 - q^2}}. \tag{7.4}$$

No gradient nonvanishing or Fréchet derivative is required.

Proof. The map $T(v) = H + G(4m|\nabla v|)$ sends the complete affine space $\ell + V$ into itself. The pointwise reverse triangle inequality and (7.1) give

$$\|T(v) - T(w)\|_V \le 4|m|C_P \, \||\nabla v| - |\nabla w|\|_2 \le q \, \|v - w\|_V .$$

Starting anywhere, its successive increments have a summable geometric bound; their limit is a fixed point. Two fixed points differ by at most q times their difference. Applying L proves equivalence to (7.2). Also $v - T(v) = G\mathcal{R}_m(v)$, so $\|u - v\|_V \le q \, \|u - v\|_V + \|\mathcal{R}_m(v)\|_{V'}$.

For the last bound let $w = u - H \in V$. Harmonicity gives $\int_D \nabla H \cdot \nabla w = 0$, while $\|w\|_V \le q \, \|\nabla u\|_2$. Thus $\|\nabla u\|_2^2 = \|\nabla H\|_2^2 + \|w\|_V^2 \le \|\nabla H\|_2^2 + q^2 \, \|\nabla u\|_2^2$. □

The smallness condition is sufficient, not necessary for scalar uniqueness. It must not be substituted for the source's all-positive-mass bounded-solution theorem. Conversely, the source's bounded solution is not automatically in this affine energy space.

Theorem 7.2 (Nonlinear flux majorant)**.** *Under Theorem 7.1, let $v \in \ell + V$ and $\sigma \in H(\mathrm{div}; D)$. Then*

$$\|\nabla(u - v)\|_2 \le \frac{\|\sigma + \nabla v\|_2 + C_P \, \|4m|\nabla v| - \mathrm{div}\, \sigma\|_2}{1 - q}. \tag{7.5}$$

If $\mathrm{div}\, \sigma = 4m|\nabla v|$, the second term vanishes. For the frozen Poisson solve $w = T(v)$ one additionally has the exact identity

$$\|\sigma + \nabla v\|_2^2 = \|\nabla(w - v)\|_2^2 + \|\sigma + \nabla w\|_2^2 . \tag{7.6}$$

The identity is for w, not directly for the nonlinear solution u.

Proof. For every zero-trace test ϕ,

$$\langle \mathcal{R}_m(v), \phi \rangle = \int_D (\nabla v + \sigma) \cdot \nabla \phi + \int_D (\mathrm{div}\, \sigma - 4m|\nabla v|)\phi.$$

Cauchy–Schwarz and Poincaré bound its dual norm by the numerator in (7.5); use (7.4). For exact equilibration, $\mathrm{div}(\sigma + \nabla w) = 0$ and $v - w \in V$. Expanding $\sigma + \nabla v = (\sigma + \nabla w) + \nabla(v - w)$ gives a zero cross term by weak integration by parts. This proves (7.6). □

This is the precise extension of the actual TR250 hypercircle proof. For piecewise polynomial v, $|\nabla v|$ need not be polynomial, so equilibration in a finite Raviart–Thomas divergence space may require a projected source and the explicitly retained mismatch. Normal flux continuity and conforming scalar trace are premises. Finite weak probes give a lower bound on a dual supremum; they cannot certify its upper bound merely by vanishing.

7.1 Wrong traces and local use

If v has the wrong trace but $d \in H^1(D)$ corrects it, apply the certificate to $v + d$, obtaining

$$\|\nabla(u - v)\|_2 \le \|\nabla d\|_2 + \frac{\|\mathcal{R}_m(v + d)\|_{V'}}{1 - q}. \tag{7.7}$$

The residual correction is at most $(1+q)\,\|\nabla d\|_2$ beyond $\|\mathcal{R}_m(v)\|_{V'}$. On a protected interior Lipschitz domain, the source solution is smooth and has a valid trace, so these bounds apply if that trace is supplied or its error is bounded. They do not magically identify its unknown boundary values from interior samples.

8 Mass, regularization, and smooth-domain perturbations

Theorem 8.1 (Finite mass and regularization budgets)**.** *On the same D with the same valid trace, let u_m, u_n be the solutions from Theorem 7.1, with $q_m, q_n < 1$. Then*

$$\|\nabla(u_m - u_n)\|_2 \le \frac{4C_P|m - n|}{1 - q_m}\,\|\nabla u_n\|_2 \le \frac{4C_P|m - n|\,\|\nabla H\|_2}{(1 - q_m)\sqrt{1 - q_n^2}}. \tag{8.1}$$

For $\varepsilon > 0$ and $q_m < 1$, there is also a unique solution $u_{m,\varepsilon} \in \ell + V$ with the right side of (7.2) *replaced by $4m\sqrt{|\nabla u|^2 + \varepsilon^2}$. It satisfies*

$$\|\nabla(u_{m,\varepsilon} - u_m)\|_2 \le \frac{4|m|C_P\varepsilon|D|^{1/2}}{1 - q_m}. \tag{8.2}$$

Proof. The residual of u_n in the m equation is $4(n-m)|\nabla u_n|$, as a weak load; its dual norm is bounded by $4C_P|m - n|\,\|\nabla u_n\|_2$. Apply (7.4) and its harmonic-lift estimate. The map $N_\varepsilon(p) = \sqrt{|p|^2 + \varepsilon^2}$ is 1-Lipschitz, so the same affine-space contraction constructs $u_{m,\varepsilon}$. Its unregularized residual is bounded in L^2 by $4|m|\varepsilon|D|^{1/2}$. Apply the same residual estimate. □

Swapping m, n supplies a second valid mass bound; use the smaller certified one. These are finite differences, not derivatives at a possibly vanishing gradient. At $m = 0$, $u_0 = H$ and the result gives an explicit weak small-mass comparison to harmonic extension, for the regularized trace setting only.

Theorem 8.2 (A smooth domain-transfer certificate)**.** *Let $T : D \to \widetilde{D}$ be an orientation-preserving C^1 bi-Lipschitz diffeomorphism with $J = \det DT > 0$. Put*

$$A = J(DT)^{-1}(DT)^{-T}, \qquad M = J(DT)^{-T}.$$

Assume $A \ge \alpha I$ almost everywhere, $\|M\|_\infty \le B$, and $a = \alpha - 4|m|C_P B > 0$. For prescribed pulled-back trace with an $H^1(D)$ lift there is a unique weak pulled-back solution U of

$$\int_D A\nabla U \cdot \nabla\phi = 4m \int_D |M\nabla U|\phi \quad (\phi \in V). \tag{8.3}$$

If v has that trace and $\mathcal{R}_m(v)$ is the flat residual in (7.3)*, then*

$$\|\nabla(U - v)\|_2 \le \frac{\|\mathcal{R}_m(v)\|_{V'} + (\|A - I\|_\infty + 4|m|C_P\,\|M - I\|_\infty)\,\|\nabla v\|_2}{a}. \tag{8.4}$$

Proof. Changing variables $y = T(x)$ in the weak physical PDE gives gradient $\nabla_y \widetilde{u} = (DT)^{-T}\nabla_x U$, Jacobian J, and (8.3); positivity of J permits $J|p| = |Jp|$. The coercive inverse for A has norm at most $1/\alpha$. Its nonlinear load is Lipschitz $V \to V'$ with constant $4|m|C_P B$. The harmonic A-lift and contraction, exactly as in Theorem 7.1, give existence and uniqueness.

Let $e = U - v \in V$ and $\mathcal{R}_A(v)$ denote the residual in (8.3). Subtract and test by e. Coercivity and the reverse triangle inequality give $a \|\nabla e\|_2 \le \|\mathcal{R}_A(v)\|_{V'}$. Finally, for $\|\phi\|_V = 1$,

$$|\langle \mathcal{R}_A(v) - \mathcal{R}_m(v), \phi\rangle| \le (\|A - I\|_\infty + 4|m|C_P \|M - I\|_\infty) \|\nabla v\|_2 .$$

Combining proves the claim. □

The theorem is a finite, quantitative version of retaining the metric, Jacobian, and boundary in the smooth transport passage of TR6. It does not differentiate a rough slit or cover a topology change or swallowing. Carathéodory convergence alone supplies none of its uniform C^1-transport or coercivity bounds. A dilation $T(x) = sx$ in two dimensions has $A = I$, $M = sI$, correctly changing the effective mass to ms; this is a useful normalization check.

9 A valid regularized Newton alternative

For a stronger local certificate, suppose D is bounded $C^{1,1}$, $p > 2$, and the fixed lift is in $W^{2,p}(D)$. Let $X = W^{2,p}(D) \cap W_0^{1,p}(D)$ with its $W^{2,p}$ norm, and let C_E bound $\|\nabla w\|_\infty \le C_E \|w\|_X$. On corrections $x \in X$ define

$$F_\varepsilon(x) = -\Delta(\ell + x) - 4m\sqrt{|\nabla(\ell + x)|^2 + \varepsilon^2} \in L^p(D). \tag{9.1}$$

For $\varepsilon > 0$ the real derivative is

$$F'_\varepsilon(x)w = -\Delta w - 4m\frac{\nabla(\ell + x)}{\sqrt{|\nabla(\ell + x)|^2 + \varepsilon^2}} \cdot \nabla w. \tag{9.2}$$

The pointwise Hessian of N_ε has operator norm at most $1/\varepsilon$. Taylor's formula therefore bounds the L^p remainder by $2|m|\varepsilon^{-1} \|\nabla w\|_\infty \|\nabla w\|_p$, proving Fréchet differentiability in these spaces. Moreover

$$\|F'_\varepsilon(x) - F'_\varepsilon(y)\|_{X \to L^p} \le L \|x - y\|_X , \qquad L = 4|m|C_E/\varepsilon. \tag{9.3}$$

No such assertion on arbitrary H^1 neighborhoods is made.

Theorem 9.1 (Local residual ball)**.** *Let $B : L^p(D) \to X$ be a bounded linear isomorphism, not merely an inverse of a retained finite matrix. At $c \in X$ suppose certified bounds give*

$$\eta \ge \|BF_\varepsilon(c)\|_X , \quad \kappa_0 \ge \|I - BF'_\varepsilon(c)\|_{X \to X} , \quad \beta \ge \|B\| .$$

If $r > 0$ satisfies

$$\eta + \kappa_0 r + \beta L r^2 \le r, \qquad q_B = \kappa_0 + \beta L r < 1, \tag{9.4}$$

then exactly one zero lies in $\overline{B}_X(c, r)$ and its distance from c is at most $\eta/(1 - q_B)$.

Proof. For $S(x) = x - BF_\varepsilon(x)$, (9.3) and integration along segments bound $\|S(x) - S(y)\|_X \le q_B \|x - y\|_X$ on the ball. Also $\|S(x) - c\| \le \eta + q_B r \le r$. Geometric summation gives a unique fixed point; injectivity of B makes it a zero of F_ε. Conversely every zero is a fixed point. Comparing that fixed point to c gives the stated error. □

This is TR1's actual `thm:ball` specialized with the explicit gradient derivative from TR6/TR8 and with all spaces fixed. At a nonvanishing-gradient neighborhood the unregularized derivative can instead be justified by a positive lower bound on $|\nabla u|$ throughout the neighborhood; the intervening gradient segments must retain that bound. A sampled nonzero gradient is insufficient.

If a regularized solution has an invertible derivative, the Banach implicit-function theorem gives its local mass derivative s for a mass-independent trace. With $a_\varepsilon(u) = \nabla u/\sqrt{|\nabla u|^2+\varepsilon^2}$, it obeys

$$(-\Delta - 4m a_\varepsilon(u)\cdot\nabla)\, s = 4\sqrt{|\nabla u|^2+\varepsilon^2}, \qquad s|_{\partial D} = 0. \tag{9.5}$$

This follows by differentiating (9.1) and retaining both its explicit mass dependence and its state derivative. The derivative inverse, the smooth trace, and $\varepsilon > 0$ are premises. The bound in (9.3) deteriorates as $\varepsilon \downarrow 0$; there is no uniform Newton-limit claim.

10 Ram panels and a finite-to-continuum checklist

Let $V_h \subset V$ be a conforming finite space with basis ϕ_j. Admissible moments $a_i = \ell_i(\sum c_j\phi_j)$ satisfy $a = Ec$. If E is nonsingular, $c = E^{-1}a$. For positive weights and all polynomial moments up to the declared degree on a nondegenerate panel, the moment Gram is positive definite because $c^T G c = \int w|\sum c_j\phi_j|^2 > 0$ for nonzero polynomial data. This is exactly the finite unisolvency passage read in revised TR250. Shared traces, not independent local jets, define the conforming space.

For a reconstruction $c = Ra$ the gradient metric becomes $K_R = R^T K R$, $K_{ij} = \int \nabla\phi_j\cdot\nabla\phi_i$. The nonlinear weak residual rows are

$$r_i(c) = \int_D \nabla(\ell + \sum_j c_j\phi_j)\cdot\nabla\phi_i - 4m\int_D |\nabla(\ell+\sum_j c_j\phi_j)|\phi_i. \tag{10.1}$$

Changing both trial and test coordinates yields $R^T r(Ra)$; it does not turn the second integral into a polynomial automatically. The physical residual certificate must include algebraic, quadrature, source-projection, geometry, conformity, boundary, and omitted-space errors in a common norm. A solution of $r_i = 0$ for finitely many i may have a nonzero continuum residual.

A practical certificate workflow is therefore: choose a regularized or protected domain and trace; reconstruct a conforming field; integrate the actual nonlinearity with an error enclosure; construct an $H(\mathrm{div})$ flux with certified divergence mismatch; verify $q < 1$ or a full Banach residual ball; and only then propagate the error to a specified output. For rough bank data, either use the bounded-residual route of Section 6 or prove an appropriate singularity-subtracted energy theory first. This is a method proposal with explicit acceptance tests, not a reported numerical solver speedup.

11 Compact drift estimates and the missing tails

At a fixed positive-mass source time, the exact physical form of (2.4) is

$$C(h) = \frac{4m}{\pi}\int_O P(Z(w))|\nabla h(w)|\,\mathrm{d}A(w). \tag{11.1}$$

Both the gradient and the Jacobian in (2.3) are needed to obtain this formula.

Theorem 11.1 (Interior drift with an explicit tail)**.** *Let $K \Subset O$ be measurable. Suppose $E_K \geq \|\nabla(h-v)\|_{L^2(K)}$ and let*

$$C_K(v) = \frac{4m}{\pi}\int_K P(Z(w))|\nabla v(w)|\,\mathrm{d}A(w), \qquad p_K = \|P\circ Z\|_{L^2(K)}.$$

Then $|C_K(h) - C_K(v)| \le (4m/\pi) p_K E_K$. *If the source's bulk estimate* $|\nabla h(w)| \le K_0 / \operatorname{crad}_O(w)$ *holds and a certified tail bound*

$$T_K \ge \int_{\mathbb{H}\setminus Z(K)} \frac{\mu(v)}{|v|^2}\, \mathrm{d}A(v) \tag{11.2}$$

is available, then

$$|C(h) - \widehat{C}_K(v)| \le \frac{4m}{\pi} p_K E_K + \frac{2K_0}{\pi} T_K + e_{\mathrm{quad}}, \tag{11.3}$$

where $e_{\mathrm{quad}} \ge |C_K(v) - \widehat{C}_K(v)|$.

Proof. Use $||p| - |q|| \le |p - q|$ and Cauchy–Schwarz on K for the first assertion. The omitted exact integral is nonnegative and bounded by $4mK_0\pi^{-1} \int_{O\setminus K} P(Z(w))/\operatorname{crad}_O(w)\, \mathrm{d}A(w)$. Under $w = \psi(v)$, conformal radius is $2\Im v|\psi'(v)|$ and area is $|\psi'(v)|^2\, \mathrm{d}A(v)$. Since $P(v) = \Im v/|v|^2$, this bound is $(2K_0/\pi) \int_{\mathbb{H}\setminus Z(K)} \mu(v)|v|^{-2}\, \mathrm{d}A(v)$. Add the compact error and quadrature error. □

The source proves K_0 depends on mass and a physical diameter bound, but does not supply a numerical value here. Neither K_0 nor T_K has been numerically enclosed in this report. Theorem 11.1 is thus an explicit conditional certificate, not a computed massive-SLE drift enclosure. Poincaré error bounds can supply E_K only with the valid local or regularized boundary realization already discussed.

Along a common prescribed evolving family, if the right side of (11.3) is a measurable e_t with $\int_0^T e_t^2\, \mathrm{d}t \le \delta^2$, then directly

$$\left\| C - \widehat{C} \right\|_{L^2(0,T)} \le \delta, \qquad \|C\|_{L^2(0,T)} \le \left\| \widehat{C} \right\|_{L^2(0,T)} + \delta. \tag{11.4}$$

If two drivers solve different feedback equations, they generally generate different domains. Applying (11.4) to one common family is not a stability theorem for those coupled curves. That further step needs law-comparison or pathwise control, stopping, domain dependence, and uniform tail estimates.

There is no global L^2 shortcut at the moving mark: even when the physical boundary is smooth near the tip, the chart weight $P(z) = \sin\theta/r$ has $\int_{0<|z|<a,\, \Im z>0} P(z)^2\, \mathrm{d}A = \infty$. Consequently an unweighted energy error alone cannot be paired with this weight by global Cauchy–Schwarz. The source's J_t and moving-mark argument address exactly the missing weighted information.

12 Exact diagnostics and their limits

The accompanying local script runs deterministic finite checks using rational and symbolic arithmetic. It is not a lattice-interface simulation, a certified PDE solve, or a continuum proof assistant. The analytic proofs above are the justification of the theorems.

1. **Contraction arithmetic.** On $(0,1)^2$, the conservative bound $C_P = 1/2$ follows by the one-dimensional zero-endpoint estimate in each vertical slice. With $m = 1/4$, $q = 1/2$, and residual upper bound $1/100$, (7.4) gives $1/50$. A putative $m = 1$ has $q = 2$ and fails this sufficient test; failure is not nonexistence.

2. **Regularization and mass budgets.** With the same mass, $|D| = 1$ and $\varepsilon = 1/1000$, (8.2) gives $1/1000$. For $m = 1/4$, $n = 1/5$, and an independently supplied $\|\nabla u_n\|_2 \le 2$, (8.1) gives $2/5$. These are exact evaluations of hypothetical certificate inputs.

3. **Nonlinear manufactured family.** For $m > 0$, $u(x, y) = (1 - e^{-4mx})/(1 - e^{-4m})$ has $u_x > 0$ and $\Delta u + 4m|\nabla u| = 0$. The top and bottom traces are the same smooth profile, not the source's Dobrushin bank jump. Symbolic differentiation verifies the equation and the tangent equation $(-\partial_x^2 - 4m\partial_x)\partial_m u = 4\partial_x u$.

4. **Exact hypercircle instance.** On the unit square take $u = x(1 - x)y(1 - y)$, $v = u/2$, $q = -\nabla u$ and $\sigma = q + (1, 0)$. Both fluxes have divergence $-\Delta u$. Exact integration gives $\|\nabla(u - v)\|_2^2 = 1/180$, $\|\sigma - q\|_2^2 = 1$, and $\|\sigma + \nabla v\|_2^2 = 181/180$. This tests signs and the linear frozen identity, not a nonlinear solution of the FK equation.

5. **Moments and shape.** The degree-two moment Gram on $[0, 1]$, $G_{ij} = 1/(i + j + 1)$, has determinant $1/2160$ and exact inverse reconstruction. A rational two-dimensional dilation verifies $A = I$, $M = sI$ in Theorem 8.2.

6. **Green and pole constants.** Symbolic differentiation of the half-plane kernel checks $\partial_t G = -(2/\pi)P(z)P(v)$ for Loewner velocities $2/z, 2/v$. Exact arithmetic solves the two pole equations to give $\kappa = 16/3$ and $\nu = (2\pi/3)C\tau$. These finite identities do not verify the source's stochastic interchanges or averaged asymptotic estimates.

All results, versions, script hashes, source hashes, and scope notes are preserved in the local evidence package. No floating-point sample is promoted to a global nonvanishing-gradient bound or a tail certificate.

13 Current primary literature and dependency limits

Public checks were performed on October 8, 2026 using only public titles, authors, and mathematical topic terms. This is a targeted primary-source search, not an exhaustive novelty or priority search. The source's exact edition remains the object of this report.

Park's massive-observable paper [2] supplies two- and four-point observable convergence and near-critical crossing estimates on angle-bounded isoradial discretizations. We inspected its scaling setup, Theorems 1.1 and 1.3, Definition 5.2, and the proof of Proposition 5.8. The last starts from square integrals of limiting fermions and uses a Green potential plus comparison. Its own introduction distinguishes observable convergence from the further analysis needed to identify a unique massive interface. M419's bounded-scalar comparison therefore has a distinct role; our report does not attribute full massive-SLE convergence to Park's 2022 paper.

Kemppainen–Smirnov [3] provides the conditional crossing framework for curve and Loewner compactness. The G2/G3 setting and main compactness statements were checked at statement level, not its entire 76-page proof. In particular, the inner-circle/boundary condition in the unforced annulus definition matters; the source separately treats bulk arms. Makarov–Smirnov [4] provides the off-critical massive-SLE program, not a blanket solution for all massive observables. The critical interface convergence paper [5] is background for $\mathrm{SLE}_{16/3}$, not an off-critical drift theorem.

A newer directly relevant primary preprint is Park's December 2025 work on Lorentz-maximal s-embeddings [6]. Its abstract, introduction, and stated setup were inspected. It treats critical non-flat maximal-surface embeddings and critical SLE convergence; that distinction prevents treating it as a replacement theorem for the fixed nonzero thermal mass of M419. No newer primary result found in the bounded search was used to certify M419's headline claim.

For numerical certification, the Prager–Synge reconstruction literature [7] gives the proper classical context for flux estimates. Its article-level abstract and scope were checked; our flux proof is supplied in full rather than imported as an unrestricted nonlinear theorem. The private Ram

reports themselves credit Lax–Milgram, contraction, and Prager–Synge. The present contribution is the specific mapping, quantitative budgets, and limitations, not a claim of originating these classical tools. Transitive proof dependencies in Park, KS, conformal boundary theory, and stochastic calculus were not exhaustively audited.

14 Assessment, advantages, and future work

14.1 What is actually gained

The all-mass estimate in Theorem 6.1 stays in the source's bounded prime-end framework and tolerates finitely many boundary jumps. Theorems 7.1–8.1 produce computable-form residual, flux, finite mass, and regularization budgets under declared smallness and lifting hypotheses. Theorems 8.2 and 9.1 isolate the actual metric and derivative assumptions for more ambitious local computations. Theorem 11.1 names the missing tail explicitly instead of claiming a PDE solve controls the full moving drift.

Advantages include exact boundary-aware reformulation; no derivative of the nonsmooth norm in the basic energy certificate; compatible moment coordinates without changing the physical trial space; and separation of source mismatch, boundary, geometry, quadrature, and tail errors. All main certificate statements have complete proofs.

Disadvantages are substantive. The bounded-residual hypothesis may fail at a singular mark and its strip constant can be large. The energy contraction can fail for a large mass or large domain, even when the scalar problem is unique. Regularizing the trace changes the problem. Newton needs a full inverse and stronger regularity; small ε worsens its constants. Rough domains and swallowing do not satisfy the smooth transport theorem automatically. A certified full drift requires K_0, the map, and its omitted tail, not just the scalar PDE approximation. No runtime, conditioning, convergence-rate, or lattice-universality improvement is measured here.

14.2 Uncompleted extensions worth pursuing

1. Prove a singularity-subtracted energy realization for the actual bank jump, including quantitative boundary remainder and rough-domain uniformity. A finite-energy correction is plausible in suitable settings but is not proved here for arbitrary prime-end boundaries.

2. Construct flux and quadrature enclosures on protected interior domains, with independently bounded artificial-boundary data, and compare their efficiency to conventional certified finite elements.

3. Obtain explicit bulk constants and certified weighted tails in (11.2), uniform along stopped families of active domains. This is the necessary next step toward numerical drift certification.

4. Develop a validated continuum inverse for the regularized linearization, including omitted modes. A retained Jacobian inverse alone cannot discharge Theorem 9.1.

5. Prove a law-level perturbation theorem for the feedback drift under common energy stops. Such a theorem must handle changing domains and measurable path functionals, not assume a common deterministic domain family. No full moving-curve limit follows from the results here.

14.3 Contribution score, separate from applicability

The internal rubric is: C1, no demonstrated contribution; C2, a little contribution, not significant; C3, notable or significant; C4, substantial or high; C5, very high with substantial significant improvements, extensions, gap closures, or important new results. Unassessed is not C1. M419 remains an **L4 applicability** topic, but this report's realized contribution is **C2/5**. The restricted certificates are useful, explicit consequences of classical arguments applied carefully to this source. They do not close a demonstrated gap in its main proof, improve its interface theorem, or establish historical novelty. Length, source reading, and the exact diagnostics are evidence quality, not additional mathematical advances.

A Local evidence and reproducibility

The source, report PDF, exact-diagnostic script and results, full-read inventory, private-passage scopes, primary-literature URLs, author audit notes, native-tool request results, build logs, every rendered page, and final hash manifest are preserved locally. Paths are relative to `C:/Users/mural/OneDrive/Documents/OpenAI-722-math`:

- `M419-Thermal-FK-Ising/TR-M419-Ram-Massive-SLE.tex` and the same stem with `.pdf`; supporting files under `evidence/`.
- `_audit/M419-author-status.txt`, `_audit/contribution-assessment-M419.json`, and `_audit/M419-author-frozen-manifest.json`.
- The working build is restricted to `_build/M419-author/`.

The exact diagnostics are reproducible with the bundled Python and `M419-Thermal-FK-Ising/evidence/check_diagnostics.py`. Native compilation was attempted but failed with a platform-directory error; the complete invocation record is retained. The on-disk PDF was exported using the existing local helper and installed compiler. Build logs, the formatting-round count, and page-render inspection records are retained. Layout verification is separate from mathematical verification. The frozen manifest excludes itself and the mutable status file.

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